

Pavan Pavithran

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Technical Skills

VAPT: Penetration testing, vulnerability assessment, network and web application security testing, security reporting and documenting

Security tools: Nessus, Burp Suite, Metasploit, Nmap, sqlmap, Splunk, Wireshark

Security Platforms/Domains: Web, Linux, Cloud, Windows, Active Directory

Networking: TCP/IP, DNS, HTTP, VPNs, firewalls, proxies

Programming & Scripting: Python, Bash, Java, C++, JavaScript

Frameworks & Standards: OWASP Top 10, CWE, MITRE ATT&CK

Certifications

EC Council – Certified Ethical Hacker (CEHv13)

June 2025

Experience

Cyber Security Intern, RedTeam Hacker Academy – Dubai, UAE

May 2025 – June 2025

- Conducted vulnerability assessments and penetration tests using Nessus, Burp Suite, and Nmap, identifying critical and high-severity vulnerabilities and supporting remediation planning.
- Prepared technical reports documenting vulnerabilities, impact, exploitation details, and remediation recommendations for client-facing security engagements.
- Monitored and investigated security alerts in Splunk, assisting with initial incident triage and analysis of suspicious activity during security operations engagements.
- Participated in client communications and recurring status meetings to discuss findings, remediation priorities, and incident-related updates.

SDE Intern, University of Illinois affiliated startup – Champaign, IL, USA

May 2024 – Aug 2024

- Engineered features for an enterprise exposure management system using Java and Spring Boot, contributing to secure backend functionality and maintaining 90%+ code coverage through testing and peer review.
- Developed and integrated REST APIs, helping improve reliability and streamline development workflows through testing and quality checks.
- Collaborated with security and developer teams to ensure robust security measures, enhancing overall system reliability and performance.

Education

University of Illinois Urbana-Champaign – Bachelor of Science in Engineering Physics

Dec 2025

Specialization: Computational Physics

Cybersecurity Projects

Vulnerability Assessment and Penetration Test – Earth (VulnHub)

drpaps.github.io/reports

Tools: Nmap, Gobuster, CyberChef, John The Ripper, Bash, Netcat, Linux privilege escalation techniques

- Conducted an end-to-end vulnerability assessment and penetration test on the VulnHub Earth machine, covering reconnaissance, service enumeration, web application assessment, exploitation, and privilege escalation to root.
- Performed network and web enumeration to map the attack surface, identifying exposed files containing cipher details, credential hints, and sensitive application data that enabled further compromise.
- Recovered administrative credentials from exposed encrypted artifacts and authenticated into the web application's admin panel, demonstrating broken access control and sensitive data exposure weaknesses.
- Achieved remote code execution by exploiting a command injection flaw in the admin interface and bypassing weak input filtering with a base64-encoded reverse shell payload, then escalated privileges to root through an insecure custom SUID binary.
- Authored a 23-page professional VAPT report documenting 6 confirmed vulnerabilities, proof-of-concept steps, impact analysis, OWASP/CWE mappings, and remediation guidance for access control, secrets management, input validation, command execution, and SUID hardening.

Python Web Reconnaissance & CTF Enumeration Utility

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Python, requests, threading, concurrent.futures, argparse, searchsploit

- Built an object-oriented Python command-line utility to automate web security reconnaissance and CTF enumeration workflows for faster target profiling and attack surface discovery.
- Implemented automated banner grabbing and web target profiling functionality using requests, enabling rapid collection of exposed technologies and infrastructure metadata during initial reconnaissance.
- Integrated SearchSploit-based local vulnerability lookups to cross-reference discovered services and technologies against publicly available exploit information during enumeration.
- Engineered a multi-threaded directory fuzzing module using threading and concurrent.futures to improve speed when enumerating hidden paths and files on web targets.
- Added dynamic wildcard calibration to reduce false positives during directory brute forcing and support more reliable content discovery across CTF and lab environments.

Security Labs & CTFs

drpaps.github.io/writeups

TryHackMe, Web Application Security, Linux/Windows Privilege Escalation, Active Directory, Cloud

- Completed 150+ hands-on security labs and CTF challenges, primarily focused on web application security testing and Linux privilege escalation, with additional exposure to Active Directory and cloud security lab environments.
- Practiced exploitation of common web vulnerabilities including broken access control, authentication weaknesses, insecure file exposure, command injection, input validation flaws, and credential abuse through guided and independent lab exercises.
- Applied tools such as Burp Suite, Nmap, sqlmap, Metasploit, Wireshark, Python, and Bash to perform enumeration, exploitation, post-exploitation, and attack-path analysis in controlled environments.
- Competed as a solo competitor in an international CTF hosted by Beagle Security and placed 15th out of 400 participants, demonstrating strong independent problem-solving in time-constrained offensive security scenarios.